

Mechanics



Lecture 2
Elementary Principles
(Goldstein Chapter 1)

What We Did Last Time

- Reviewed basic principles of Newtonian Mechanics
 - Define standard notations and usages
 - Momenta, conservation laws, kinetic & potential energies
- Concentrated on the motion of a single particle

Goals for Today

- Single → multi-particle system
 - Force between particles
 - Laws of action and reaction ← Newton's 3rd Law
- Introduce constraints
 - Holonomic and nonholonomic constraints
- Introduce Lagrange's Equation

System of Particles

- More than one particles? → Just add indices!

$$\mathbf{F}_i = \dot{\mathbf{p}}_i$$

$$\mathbf{N}_i = \dot{\mathbf{L}}_i$$

- Subtlety: **F** may be working between particles
 - Distinguish between **internal** and **external** forces

$$\mathbf{F}_i = \sum_j \mathbf{F}_{ji} + \mathbf{F}_i^{(e)}$$

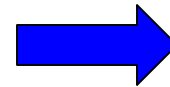
The diagram illustrates the components of the force equation. A central blue box contains the equation $\mathbf{F}_i = \sum_j \mathbf{F}_{ji} + \mathbf{F}_i^{(e)}$. Three yellow boxes with red borders provide explanations: 'Force acting on particle i ' has an arrow pointing to $\mathbf{F}_i$; 'Force from particle j ' has an arrow pointing to $\mathbf{F}_{ji}$; and 'Force from outside' has an arrow pointing to $\mathbf{F}_i^{(e)}$.

- Now add up over i to see the overall picture

Sum of Particles

$$\sum_i \mathbf{F}_i = \sum_{\substack{i,j \\ i \neq j}} \mathbf{F}_{ji} + \sum_i \mathbf{F}_i^{(e)} = \sum_{i < j} (\mathbf{F}_{ji} + \mathbf{F}_{ij}) + \sum_i \mathbf{F}_i^{(e)}$$

- This term vanishes if $\mathbf{F}_{ij} = -\mathbf{F}_{ji}$
 - Weak law of action and reaction

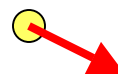


$$\sum_i \mathbf{F}_i = \sum_i \mathbf{F}_i^{(e)}$$

Forces two particle exert on each other are equal and opposite

- C.f. the strong law of action and reaction

Forces two particle exert on each other are equal, opposite, and along the line joining the particles



Sum of Particles

- Now consider the equations of motion

$$\sum_i \mathbf{F}_i = \sum_i \mathbf{F}_i^{(e)} = \sum_i \dot{\mathbf{p}}_i = \frac{d^2}{dt^2} \sum_i m_i \mathbf{r}_i$$

- Define **center of mass** $\mathbf{R} \equiv \frac{\sum m_i \mathbf{r}_i}{\sum m_i} = \frac{\sum m_i \mathbf{r}_i}{M}$


$$M\ddot{\mathbf{R}} = \sum_i \mathbf{F}_i^{(e)} \equiv \mathbf{F}^{(e)}$$

Center of mass moves like a particle of mass M under total external force $\mathbf{F}^{(e)}$

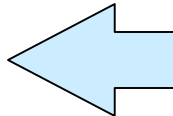
Total Linear Momentum

- The sum of the linear momenta is

$$\mathbf{P} = \sum_i \mathbf{p}_i = \sum_i m_i \dot{\mathbf{r}}_i = M \dot{\mathbf{R}}$$

- Taking the time derivative

$$\dot{\mathbf{P}} = M \ddot{\mathbf{R}} = \mathbf{F}^{(e)}$$



Newton's equation
of motion for the
center of mass

- Conservation of total linear momentum

If the total external force $\mathbf{F}^{(e)}$ is zero, the total linear momentum $\mathbf{P}$ is conserved

Assumed weak law of action & reaction

Total Angular Momentum

- The sum of the angular momenta is

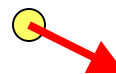
$$\mathbf{L} = \sum_i \mathbf{L}_i = \sum_i \mathbf{r}_i \times \mathbf{p}_i$$

- Take time derivative and use $\dot{\mathbf{p}}_i = \mathbf{F}_i = \sum_j \mathbf{F}_{ji} + \mathbf{F}_i^{(e)}$

$$\dot{\mathbf{L}} = \sum_{\substack{i,j \\ i \neq j}} \mathbf{r}_i \times \mathbf{F}_{ji} + \sum_i \mathbf{r}_i \times \mathbf{F}_i^{(e)}$$

Total external
torque

- This term vanishes only if $\mathbf{F}_{ji}$ satisfies the strong law of action and reaction



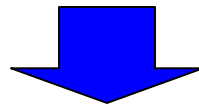
Total Angular Momentum

- Assuming strong law of action and reaction

$$\dot{\mathbf{L}} = \sum_i \mathbf{r}_i \times \mathbf{F}_i^{(e)} = \sum_i \mathbf{N}_i^{(e)} = \mathbf{N}^{(e)}$$

→ Conservation of total angular momentum

If the total external torque $\mathbf{N}^{(e)}$ is zero, the total angular momentum $\mathbf{L}$ is conserved

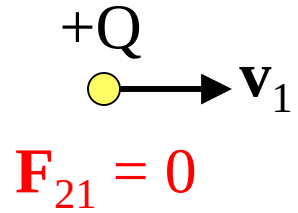


A multi-particle system (= extended object) can be treated as if it were a single particle if the internal forces obey the strong law of action and reaction

Laws of Action and Reaction

- Most forces we know obey strong law of action and reaction

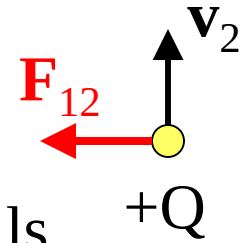
- Gravity, electrostatic force



- There are rare exceptions

- E.g. Lorenz force felt by moving charges

- Conservation of linear & angular momenta fails



- Take into account the EM field

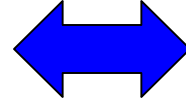
- Particles exchange forces with the field

- The field itself has linear & angular momenta

→ Conservation laws restored

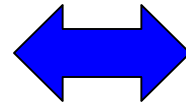
Conservation Laws

Weak law of action-reaction



Conservation of **P**

Strong law of action-reaction



Conservation of **L**

- We will see (in 2 lectures) that **P** and **L** must be conserved if the laws of physics are isotropic in space
 - No special origin
 - No special orientation
- If we accept these symmetries as fundamental principles, all forces must satisfy the action-reaction laws → “Proof” of Newton’s 3rd Law

Total Angular Momentum

- Define particle i 's position from the center of mass

$$\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}$$

- Also define the velocities $\mathbf{v}'_i = \dot{\mathbf{r}}'_i$ $\mathbf{v} = \dot{\mathbf{R}}$
- Calculate the total angular momentum

$$\mathbf{L} = \sum_i \mathbf{r}_i \times \mathbf{p}_i = \sum_i (\mathbf{r}'_i + \mathbf{R}) \times m_i (\mathbf{v}'_i + \mathbf{v})$$


$$\mathbf{L} = \mathbf{R} \times M\mathbf{v} + \sum_i \mathbf{r}'_i \times m_i \mathbf{v}'_i$$

Angular momentum of motion concentrated at the center of mass

Angular momentum of motion around the center of mass

Kinetic Energy

- The work done by force $W_{12} = \sum_i \int_1^2 \mathbf{F}_i \cdot d\mathbf{s}_i$
 - Positions 1 and 2 are now **configurations** (sets of positions)
- Use equations of motion to derive

$$W_{12} = T_2 - T_1 \quad \text{where} \quad T = \sum_i \frac{1}{2} m_i v_i^2$$

- One can split T into two pieces

$$T = \sum_i \frac{1}{2} m_i (\mathbf{v} + \mathbf{v}'_i) \cdot (\mathbf{v} + \mathbf{v}'_i) \rightarrow \frac{1}{2} M v^2 + \sum_i \frac{1}{2} m_i v_i'^2$$

Motion concentrated at
the center of mass

Motion around the
center of mass

Potential Energy

$$\nabla_i \equiv \left(\frac{\partial}{\partial x_i} \quad \frac{\partial}{\partial y_i} \quad \frac{\partial}{\partial z_i} \right)$$

- Assume conservative external force $\mathbf{F}_i^{(e)} = -\nabla_i V_i$

$$\sum_i \int_1^2 \mathbf{F}_i^{(e)} \cdot d\mathbf{s}_i = -\sum_i \int_1^2 \nabla_i V_i \cdot d\mathbf{s}_i = -\sum_i V_i \Big|_1^2$$

- Assume also conservative internal forces $\mathbf{F}_{ji} = -\nabla_i V_{ij}$

- To satisfy strong law of action/reaction

$$V_{ij} = V_{ij}(|\mathbf{r}_i - \mathbf{r}_j|)$$

← Potential depends only on the distance

$$\sum_{\substack{i,j \\ i \neq j}} \int_1^2 \mathbf{F}_{ji} \cdot d\mathbf{s}_i = -\sum_{\substack{i,j \\ i \neq j}} \int_1^2 \nabla_i V_{ij} \cdot d\mathbf{s}_i$$

→ Bit of work

$$-\frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} V_{ij} \Big|_1^2$$

Energy Conservation

- If all forces are conservative, one can define total potential energy:

$$V = \sum_i V_i + \frac{1}{2} \sum_{\substack{i,j \\ i \neq j}} V_{ij}$$

- Then the total energy $T + V$ is conserved
- The second term is **internal potential energy**
 - It depends on the distances between all pairs of particles
 - Constant if particles' relative configuration is fixed
 - **Rigid bodies**

Constraints

- Equation of motion $m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i = \mathbf{F}_i^{(e)} + \sum_j \mathbf{F}_{ji}$ assumes that particles can move anywhere in space
 - Not generally true
 - In fact never true – Free space is an idealization
 - Amusement-park ride constrained (hopefully) on a rail
 - Billiard balls on a pool table
- How can we accommodate constraints in the equation of motion?
 - Depends on the type of the constraint

Holonomic Constraints

- Constraints may be expressed by

$$f(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots, t) = 0$$

A holonomic
constraint

- Particle on the x-y plane $z = 0$
- Rigid body $(\mathbf{r}_i - \mathbf{r}_j)^2 - c_{ij}^2 = 0$
- All other cases are called **nonholonomic**
 - It means “we don’t really want to mess with it”
 - May be inequalities such as $z \geq 0$
 - May depend on derivatives such as $\dot{\mathbf{r}}_i$
- We will deal only with holonomic constraints

Independent Variables

- A holonomic constraint reduces the number of independent variables by 1
 - If $z = 0$, you are left with only x and y
 - You may be able to solve the constraint for one variable

$$f(\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots, t) = 0 \longrightarrow x_1 = g(y_1, z_1, \mathbf{r}_2, \mathbf{r}_3, \dots, t)$$

- Then you can drop this variable
 - You may have to switch to a different set of variables
 - For a particle on a sphere $x^2 + y^2 + z^2 = c^2$
a good choice is (θ, ϕ)
- New set of variables $\rightarrow$ Generalized coordinates

Generalized Coordinates

- N particles have $3N$ degrees of freedom
 - Introducing k holonomic constraints reduces it to $3N - k$
 - Using generalized coordinates $q_1, q_2, \dots, q_{3N-k}$

$$\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, q_3, \dots, t)$$

Transformation equations from $(\mathbf{r}_l)$ to (q_l)

- Example:

$$\begin{cases} x = c \sin \theta \cos \phi \\ y = c \sin \theta \sin \phi \\ z = c \cos \theta \end{cases}$$

Transformation
from (x, y, z)
to (θ, ϕ)

Now What?

- We know the equations of motion for ($\mathbf{r}_i$)

$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i = \mathbf{F}_i^{(e)} + \sum_j \mathbf{F}_{ji}$$

- We know how to include constraints by switching to generalized coordinates

$$\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, q_3, \dots, t)$$

- How can we transform the equation of motion to the generalized coordinates?



Lagrange's Equations

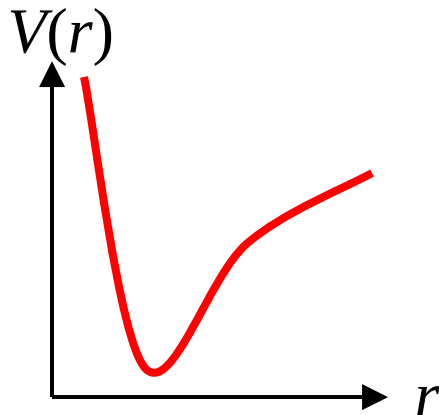
Why Constraints?

- Constraint is an idealized classical concept
 - Nothing is perfectly constrained in QM Uncertainty
- How useful is it to switch between coordinates?

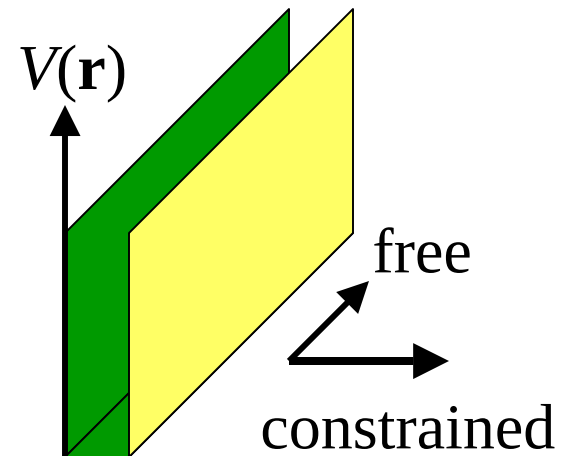
More than it seems

Constraints and Force

- A holonomic constraint is an infinitely strong force
 - Or an infinitely high potential wall
- Reality is always smoother
 - E.g. electron of a hydrogen atom



It's still true that the electron feels **strong radial (binding) force**, while it can move **freely around the nucleus**



Binding force makes the **r** direction **special**

Force and Symmetry

- Without forces, all coordinate systems are equal
 - x-y-z system is the simplest
- Forces break the symmetry
 - Some coordinate system works better than others
- Generalized coordinates offer natural way of handling systems with such forces
- Constraints are extreme cases
 - We develop our technique with them

OK, back to the business...

Lagrange's Equations

Recipe

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0$$

$$L(q, \dot{q}, t) \equiv T - V$$

Kinetic energy

Potential energy

Lagrangian

- Express $L = T - V$ in terms of generalized coordinates $\{q_j\}$, their time-derivatives $\{\dot{q}_j\}$, and time t
 - The potential $V = V(q, t)$ must exist
 - i.e. all forces must be conservative
- Let's do a quick example to see how it works

Ex: Particle on a Line

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_j} \right) - \frac{\partial L}{\partial q_j} = 0$$

- A particle moving on the x -axis $x = x(t), y = 0, z = 0$

- Kinetic and potential energies:

$$T = \frac{m}{2} \dot{x}^2$$

$$V = V(x)$$



$$L = \frac{m}{2} \dot{x}^2 - V(x)$$

Lagrange's Eqn

$$m\ddot{x} + \frac{\partial V}{\partial x} = 0$$

- Equivalent to Newton's Eqn given that

$$F_x = -\frac{\partial V}{\partial x}$$

OK, it works

Summary

- Discussed multi-particle systems
 - Internal and external forces
 - Laws of action and reaction
 - Momenta, conservation laws, kinetic & potential energies
- Introduced constraints
 - Holonomic and nonholonomic constraints
 - Generalized coordinates
- Introduced Lagrange's Equations
 - Next: Prove that Lagrange's and Newton's Equations are equivalent